ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. ME ENGINEERING — SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain Sand Casting process: Pattern allowances, Molding sand properties, Gating system, and Risers. [14 Marks]

[OR]

Q.1 (Alt) Describe Special Casting methods: Die Casting, Investment Casting, and Centrifugal Casting. [14 Marks]

--- UNIT II ---

Q.2 Explain Arc Welding processes: SMAW, TIG (GTAW), and MIG (GMAW). Explain HAZ (Heat Affected Zone). [14 Marks]

[OR]

Q.2 (Alt) Describe Resistance Welding (Spot, Seam, Projection) and Solid-State welding (Friction welding). [14 Marks]

--- UNIT III ---

Q.3 Explain Metal Forming: Hot working vs Cold working. Describe Rolling process and Angle of Bite. [14 Marks]

[OR]

Q.3 (Alt) Describe Forging (Open die, Closed die, Drop forging) and Extrusion (Direct and Indirect). [14 Marks]

--- UNIT IV ---

Q.4 Explain Sheet Metal operations: Blanking, Piercing, Bending, Drawing, and Clearance calculation. [14 Marks]

[OR]

Q.4 (Alt) Describe Powder Metallurgy process: Powder production, Blending, Compacting, and Sintering. [14 Marks]

--- UNIT V ---

Q.5 Explain Mechanics of Metal Cutting: Merchant's Circle Diagram and cutting force calculations. [14 Marks]

[OR]

Q.5 (Alt) Describe Tool Wear mechanisms, Taylor's Tool Life Equation, and Cutting Fluids. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****